

Question 3

(a)

Using the Orthogonality Principle for an anti-causal filter:

$$\mathbf{E}\{e[n]x^*[n-l]\} = 0 \quad \text{for } l \in [-p, 0] \quad [1 \text{ mark}]$$

Substitute the error signal $e[n] = d[n] - \hat{d}[n]$:

$$\mathbf{E}\left\{\left(d[n] - \sum_{k=-p}^0 h[k]x[n-k]\right)x^*[n-l]\right\} = 0 \quad [1 \text{ mark}]$$

Expanding and writing in terms of correlation sequences:

$$r_{dx}[l] = \sum_{k=-p}^0 h[k]r_{xx}[l-k] \quad [1 \text{ mark}]$$

Given $x[n] = d[n] + v[n]$, and $d[n], v[n]$ are uncorrelated, we find r_{dx} and r_{xx} :

$$r_{dx}[l] = \mathbf{E}\{d[n](d[n-l] + v[n-l])^*\} = r_{dd}[l] \quad [1 \text{ mark}]$$

$$r_{xx}[l-k] = \mathbf{E}\{(d[n-k] + v[n-k])(d[n-l] + v[n-l])^*\} = r_{dd}[l-k] + r_{vv}[l-k] \quad [1 \text{ mark}]$$

Substituting these back yields the Wiener-Hopf equations:

$$\sum_{k=-p}^0 h[k](r_{dd}[l-k] + r_{vv}[l-k]) = r_{dd}[l] \quad \text{for } l \in [-p, 0] \quad [1 \text{ mark}]$$

(b)

For a causal FIR Wiener filter of the same length p :

$$\sum_{k=0}^p h_{\text{causal}}[k](r_{dd}[l-k] + r_{vv}[l-k]) = r_{dd}[l] \quad \text{for } l \in [0, p] \quad [1 \text{ mark}]$$

The structure of the equations is identical, but the indices are flipped in time (anti-causal uses $-p$ to 0, causal uses 0 to p).

$$h_{\text{causal}}[k] = h[-k] \quad , \quad \forall k \in \{0, \dots, p\} \quad [1 \text{ mark}]$$

(c)

For $p = 2$, $r_d[k] = (0.2)^{|k|}$, and $r_v[k] = \delta[k]$, the input auto-correlation $r_{xx}[k] = r_d[k] + r_v[k]$ is:

$$r_{xx}[0] = 2, \quad r_{xx}[\pm 1] = 0.2, \quad r_{xx}[\pm 2] = 0.04 \quad [1 \text{ mark}]$$

Evaluating the summation from part (a) at lags $l = -2, -1, 0$:

$$2h[-2] + 0.2h[-1] + 0.04h[0] = 0.04 \quad [1 \text{ mark}]$$

$$0.2h[-2] + 2h[-1] + 0.2h[0] = 0.2 \quad [1 \text{ mark}]$$

$$0.04h[-2] + 0.2h[-1] + 2h[0] = 1 \quad [1 \text{ mark}]$$